

Functional Outcomes After Intensive Blood Pressure Reduction in Deep and Lobar Intracerebral Hemorrhage

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Abstract

Background and Objectives

Intensive blood pressure (BP) reduction may reduce the risk of poor functional outcomes in patients with acute intracerebral hemorrhage (ICH). Whether these potential benefits apply similarly to lobar and deep ICH, 2 biologically and clinically distinct subtypes, remains uncertain. We tested the hypothesis that intensive BP reduction has differential effects on functional outcomes based on ICH location.

Methods

We performed a stepwise meta-analysis, stratified by ICH location, of 3 landmark randomized clinical trials of intensive BP reduction: Acute Cerebral Hemorrhage (ATACH-2), Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 2 (INTERACT2), and INTERACT3. Step 1 pooled ATACH-2 and INTERACT2 (3-month outcomes; intensive BP lowering only). Step 2 added INTERACT3 (6-month outcomes; care bundle protocol with intensive BP lowering plus glucose management, antipyresis, and anticoagulant reversal). For ATACH-2, we used pooled results from individual patient data (adjusted for age, Glasgow Coma Scale score, and presence of intraventricular hemorrhage), whereas for INTERACT2 and INTERACT3, we used their pooled, publicly available results. Our exposure of interest contrasted intensive systolic BP targets <140 mm Hg vs standard care 140–180 mm Hg. Our outcome of interest was poor functional outcome, defined as a modified Rankin Scale score of 4–6 in ATACH-2 and INTERACT3 and of 3–6 in INTERACT2.

Results

Step 1 included a total of 2,983 patients with deep and 537 patients with lobar ICH (mean age 63 y/o, 37% female). Intensive BP reduction was not associated with a significant difference in poor functional outcome for either deep (odds ratio [OR] 0.89; 95% CI 0.40–1.98; $I^2 = 0\%$) or lobar (OR 0.92; 95% CI 0.73–1.17; $I^2 = 0\%$) ICH. Step 2 included a total of 7,917 patients with deep and 1,105 patients with lobar ICH (mean age 63 y/o, 37% female). Similarly, intensive BP reduction was not associated with a significant difference in poor outcome in both deep (OR 0.82; 95% CI 0.57–1.18; $I^2 = 60\%$) and lobar (OR 0.97; 95% CI 0.76–1.24; $I^2 = 0\%$) ICH.

Discussion

In this stepwise meta-analysis of 3 landmark ICH trials, intensive BP reduction did not demonstrate a significant benefit in either deep or lobar ICH. Although our estimates did not reach statistical significance, the direction of effect in deep ICH and the substantial heterogeneity across trials, particularly with the inclusion of INTERACT3, limit firm conclusions. Given these uncertainties and the biological distinctions between deep and lobar ICH, future well-powered studies specifically designed to test whether intensive BP reduction has differential effects by hematoma location are warranted.

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Glossary

ATACH-2 = acute cerebral hemorrhage; **BP** = blood pressure; **GCS** = Glasgow Coma Scale; **ICH** = intracerebral hemorrhage; **INTERACT2** = Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial; **mRS** = modified Rankin Scale; **OR** = odds ratio.

Introduction

Acute spontaneous intracerebral hemorrhage (ICH) is the most serious and least treatable form of stroke, with mortality higher than 40% in the first 30 days and 50% of survivors experiencing permanent disability.^{1,2} Elevated blood pressure (BP) is present in most acute ICH cases and is associated with hematoma expansion and poor functional outcomes.^{3,4}

Four landmark randomized clinical trials have studied the impact of intensive BP reduction in acute ICH with mixed results. In the second Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial (INTERACT2)⁵ and the second Intensive Blood-Pressure Lowering in Patients with Acute Cerebral Hemorrhage (ATACH-2)⁶ clinical trials, intensive BP reduction did not result in a significant reduction of death or severe disability at 3 months. The third INTERACT3⁷ showed that a care bundle protocol, which included intensive BP reduction as the main therapeutic strategy, improved functional recovery at 6 months. Finally, INTERACT4 showed that ultra-early BP reduction in the prehospital setting is beneficial for patients with ICH.⁸ However, patients with lobar ICH were underrepresented in ATACH-2, INTERACT2, and INTERACT3 trials (<18%), whereas INTERACT4 did not report ICH location.⁵⁻⁸ Recent meta-analyses pooling the effects of these trials have shown that intensive BP reduction is associated with improved functional outcomes in acute ICH, reinforcing its therapeutic potential at the population level.^{9,10}

ICH is commonly classified into deep and lobar locations, which represent biologically distinct entities with different underlying mechanisms. Deep ICH is primarily driven by chronic hypertension, which leads to structural damage and remodeling of small perforating arteries.^{11,12} By contrast, lobar ICH correlates with cerebral amyloid angiopathy and associated pathologies.^{11,12} One key pathophysiologic feature of ICH is hematoma expansion, which varies in frequency, timing, and clinical consequences between deep and lobar hemorrhages.¹³⁻¹⁵ This is important because hematoma expansion is one of the main targets of intensive BP reduction, yet it remains unclear whether this therapeutic approach benefits patients with both lobar and deep ICH equally. Indeed, in a recent exploratory analysis of the ATACH-2 trial, it was found that intensive BP reduction reduced hematoma expansion in deep but not in lobar ICH, without a significant effect on 90-day functional outcomes in either group, highlighting the need for more well-powered studies to determine whether intensive BP reduction provides functional benefits across both ICH subtypes.¹⁶

Thus, we conducted a meta-analysis of the landmark ATACH-2, INTERACT2, and INTERACT3 randomized clinical trials to test the hypothesis that intensive BP reduction has differential effects on functional outcomes depending on deep or lobar ICH location.

Methods

Study Design

We performed a stepwise meta-analysis using aggregate data from the landmark ATACH-2, INTERACT2, and INTERACT3 international, multicenter, randomized clinical trials, stratified by deep and lobar ICH.⁵⁻⁷ We chose these 3 studies because, according to previous literature reviews, they are the only ICH randomized clinical trials investigating intensive BP reduction and presenting functional outcomes by location.¹¹ INTERACT4 was not included because no publicly available data were reported stratified by ICH location.⁸ Details about the design and inclusion/exclusion criteria of these studies are listed in Table 1. We included all patients with available neuroimaging, covariate, and outcome data for analysis from ATACH-2, and all published location-stratified summary results from INTERACT2 and INTERACT3. In step 1, we meta-analyzed the results of ATACH-2 and INTERACT2 as both studies assessed functional outcomes at 3 months, with intensive BP lowering as the sole intervention. In step 2, we incorporated the aggregate-level 6-month functional outcome results of INTERACT3, where BP reduction formed the core of the intensive care bundle and was the principal driver of its beneficial effect.¹⁷ Each study was approved by the appropriate institutional review board, and written informed consent was obtained from each participant or legal representative. For ATACH-2, we accessed a publicly available, deidentified version of the study's data set featuring individual patient data, through the National Institute of Neurological Disorders and Stroke Archive of Randomized Clinical Trials. For INTERACT2 and INTERACT3, we did not have access to individual patient data and used instead their pooled and publicly available results. Reporting followed established recommendations for meta-analyses, including relevant items from Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020, adapted for this targeted analysis of prespecified randomized clinical trials.¹⁸

Neuroimaging Analyses and Bleeding Location Ascertainment

Neuroimaging analysis was performed in image analysis centers across all studies, with readers blinded to treatment assignments and clinical findings. ICHs were classified according to their location as lobar or deep. Because ATACH-2 classified the locations of participants' ICH as either lobar,

Table 1 Comparison of the Baseline Characteristics of the Antihypertensive Treatment of Acute Cerebral Hemorrhage-2 (ATACH-2), Intensive Care Bundle With Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 2 (INTERACT2), and INTERACT3 Randomized Clinical Trials

	ATACH-2	INTERACT2	INTERACT3
Study design characteristics			
Type of Trial	International, randomized, multicenter, two-group, open-label clinical trial	International, randomized, multicenter, open-label, blinded end point, clinical trial	International, multicenter, stepped wedge, cluster randomized, blinded, clinical trial
Number of countries involved	6	18	10
Inclusion criteria	≥18 years old with CT-confirmed ICH, GCS score ≥5 with at least 1 systolic BP reading ≥180 mm Hg within 4.5 h of symptom onset	≥18 years old with CT-confirmed ICH, GCS score ≥6, with systolic BP ≥150 and ≤220 within 6 h of symptom onset	≥18 years old with CT-confirmed ICH within 6 h of symptom onset
Intervention	Systolic BP reduction to <140 mm Hg within 1 h of randomization with IV nicardipine	Systolic BP reduction to <140 mm Hg within 1 h of randomization with any oral or IV antihypertensives according to site availability	Intensive care bundle ^a , including systolic BP reduction to <140 mm Hg within 1 h of randomization with any oral or IV antihypertensive
Comparator	Conventional BP lowering to 140–180 mm Hg with IV nicardipine	Conventional BP lowering to 140–180 mm Hg with any oral or IV antihypertensives according to site availability	Usual care ^b
Primary outcome	3-mo mRS score of 4–6 (dichotomous)	3-mo mRS score of 3–6 (dichotomous)	6-mo mRS score (ordinal)
Year published	2016	2013	2023
Participant characteristics			
Demographics			
Randomized, n	1,000	2,839	7,036
Age, mean (SD)	61.9 (13)	63.5 (12.9)	62 (12.6)
Female sex, n (%)	380 (38)	1,053 (37.1)	2,533 (36)
Asian ethnicity, n (%)	562 (56.2)	1,919 (67.8)	6,350 (90.3)
Baseline neuroimaging and clinical characteristics			
ICH location, n (%)^c			
Deep	833 (89)	2,150 (83)	4,934 (90)
Lobar	104 (11)	433 (17)	568 (10)
ICH volume (cm³), median (IQR)	10.3 (1.6–82)	11 (6–19)	15 (7.5–30)
Presence of IVH, n (%)	265 (26.8)	739 (28.3)	2,034 (29.7)
Baseline systolic BP, mean (SD)	200.6 (27)	179 (17)	174.5 (28.3)

Abbreviations: BP = blood pressure; GCS = Glasgow Coma Scale; ICH = intracerebral hemorrhage; IQR = interquartile range; IVH = intraventricular hemorrhage; mm Hg = millimeters of mercury.

^a Other interventions of the care bundle included blood glucose control in hyperglycemia, temperature control in hyperpyrexia, and anticoagulant reversal. ^b Where location of care, its delivery, monitoring, and treatments were made by the treating clinical team.

^c A total of 63, 256 and 1,534 participants from the ATACH-2, INTERACT2, and INTERACT3 trials had no data reported or available for ICH location analysis.

thalamic, or basal ganglia, we considered all thalamic and basal ganglia bleeds as deep ICH.

Exposure of Interest

Our exposure of interest was conventional vs intensive BP lowering as defined and evaluated in each trial (Table 1).^{5–7} In all trials, intensive BP lowering was <140 millimeters of mercury (mm Hg) while conventional BP lowering was 140–180 mm Hg.

INTERACT3 also compared intensive and conventional BP lowering but incorporated additional components into a broader care bundle, including antipyretic therapy, glucose management, and anticoagulant reversal.⁷

Outcomes of Interest

Functional outcomes were assessed using the modified Rankin Scale (mRS) at 3 months in INTERACT2 and ATACH-2

and at 6 months in INTERACT3, as a dichotomous outcome (good vs poor), with evaluations conducted by trial investigators who were blinded to treatment assignment. Poor outcome was defined as mRS scores 4–6 in ATACH2 and INTERACT3 and mRS scores 3–6 in INTERACT2. Owing to the use of published aggregate data from INTERACT2 and INTERACT3, we maintained the original outcome definitions and time points used in each trial, resulting in minor differences in the mRS thresholds and follow-up durations.

Statistical Analysis

We performed meta-analyses using aggregate-level data for both deep and lobar ICH. Because ATACH-2 did not provide aggregate-level results for deep ICH, we used its individual-level patient data and fit a multivariable logistic regression model to obtain these results, subsequently including them in the meta-analysis. This model was adjusted for age, baseline Glasgow Coma Scale (GCS), and presence/absence of intraventricular hemorrhage, the same covariates used in the original ATACH-2 analysis.⁶ Because INTERACT3 did not report location-stratified data using a dichotomous mRS, we first calculated the corresponding odds ratios (ORs) using a 2×2 table (eTable 1) and then used these for meta-analysis. We used the Q test and I^2 to assess total heterogeneity. We used random effects with inverse variance-weighted meta-analyses to pool study-specific results because of the existing methodological and clinical heterogeneity across these studies. We also used Baujat plots to understand how each study contributed to and influenced overall heterogeneity and results, respectively. Finally, we conducted 3 separate sensitivity analyses. First, we report the results of steps 1 and 2 using fixed-effects estimates to compare them with our random-effects estimates. Second, we meta-analyzed the results of ATACH-2 and INTERACT3 using an ordinal definition of the mRS by first fitting a multivariable ordinal linear regression model adjusted for age, baseline GCS score, and presence/absence of intraventricular hemorrhage, using individual-level patient data for deep and lobar ICH in ATACH-2 and then meta-analyzed these pooled results with the stratified ordinal mRS results reported in INTERACT3. We were not able to include INTERACT2 in this sensitivity analysis because they did not report ordinal mRS results stratified by ICH location. Third, because the published results of INTERACT2 and INTERACT3 were unadjusted and those of ATACH-2 were not, we repeated steps 1 and 2 of our analysis using the results of unadjusted logistic regressions for deep and lobar ICH in ATACH-2. The threshold for significance was set at a 2-sided p value of 0.05. We conducted all analyses between July 2023 and October 2024 using the R statistical package version 4.1.1 and the meta and Modern Applied Statistics packages for meta-analysis, drawing forest plots and ordinal regressions, respectively.^{19–21} All figures were generated with BioRender (Falcone, G., 2025).²²

Finally, risk of bias for each trial was assessed using the Cochrane Risk of Bias 2.0 tool for individually randomized trials.²³ The tool considers 5 domains: (D1) bias arising from

the randomization process; (D2) bias due to deviations from intended interventions; (D3) bias due to missing outcome data; (D4) bias in measurement of the outcome; and (D5) bias in selection of the reported result. For each domain, we judged risk of bias as “low” or “some concerns” and derived an overall risk-of-bias judgment per trial. Two reviewers (SCT and MAH) independently performed the assessment, with disagreements resolved by discussion.

Standard Protocol Approvals, Registrations, and Patient Consents

ATACH-2, INTERACT2, and INTERACT3 were registered in ClinicalTrials.gov (unique identifiers: NCT01176565, NCT00716079, and NCT03209258, respectively). The respective institutional review boards or ethics committees at each participating site for each trial approved the study protocols, and all participants or their legally authorized representatives provided written informed consent.

Data Availability

For ATACH-2, we accessed a publicly available, deidentified version of the study’s data set featuring individual patient data through the National Institute of Neurological Disorders and Stroke Archive of Randomized Clinical Trials. For INTERACT2 and INTERACT3, we used the publicly available, published results of the respective publications.^{5,7}

Results

The 3 trials combined randomized a total of 10,875 participants; 37% were female, and 81% were Asian. Among the 9,022 individuals with available data, 7,917 patients (88%) had deep ICH and 1,105 patients (12%) had lobar ICH (Figure 1). Table 1 provides a comparison of the ATACH-2, INTERACT2, and INTERACT3 studies, including the baseline characteristics of the participants enrolled in each study.

Step 1: Intensive BP Reduction Strategies in Deep and Lobar ICH in ATACH-2 and INTERACT2

Step 1 included² 983 patients with deep ICH and 537 patients with lobar ICH (mean age 63 years old, 37% female individuals). Intensive BP reduction was not associated with a significant difference in functional outcome for both patients with deep (OR 0.89; 95% CI 0.40–1.98, I^2 0%, Figure 2, upper-left panel) and lobar (OR 0.92; 95% CI 0.73–1.17, I^2 0%, Figure 2, lower-left panel) ICH. Baujat plots confirmed that INTERACT2 had a higher influence on overall results, and both studies had a low contribution to overall heterogeneity (eFigure 1).

Step 2: Intensive BP Reduction Strategies in Deep and Lobar ICH in ATACH-2, INTERACT2, and INTERACT3

After including INTERACT3 in the analysis, there were 7,917 patients with deep ICH and 1,105 patients with lobar ICH (mean age 63 years, 37% female). Similarly to step 1, we found no statistically significant difference in functional outcomes after

intensive BP reduction in both deep ICH (OR 0.82; 95% CI 0.57–1.18, I^2 60%, Figure 2, upper-right panel) and lobar ICH (OR 0.97; 95% CI 0.76–1.24, I^2 0%, Figure 2, lower-right panel). Baujat plots showed that INTERACT3 exerted the greatest influence on the pooled estimates for deep ICH, whereas INTERACT2 contributed most to the overall estimates for lobar ICH. For deep ICH, INTERACT3 also accounted for a larger proportion of the between-study heterogeneity; however, for both deep and lobar ICH, all 3 trials demonstrated only minimal contributions to overall heterogeneity (eFigure 2).

Sensitivity Analyses: Fixed-Effects Meta-Analyses, Ordinal mRS Outcome, and Unadjusted Results in ATACH-2

In the first sensitivity analysis, we repeated steps 1 and 2, including the same number of patients for each study, using fixed-effects, inverse variance–weighted meta-analysis instead of the random-effects model. For step 1, results remained consistent, showing no difference in functional outcomes after intensive BP reduction for either deep or lobar ICH (Figure 3, left panels). However, in step 2, intensive BP reduction showed a significant benefit in functional outcomes for patients with deep ICH (OR 0.79, 95% CI 0.72–0.86, I^2 60%, Figure 3, upper-right panel), in contrast to patients with lobar ICH who showed no significant benefit (OR 0.97, 95% CI 0.80–1.18, I^2 0%, Figure 3, lower-right panel). In the second sensitivity analysis, we included 5,767 patients with deep ICH and 672 patients with lobar ICH in ATACH-2 and INTERACT3 to evaluate whether the observed results remained consistent when using an ordinal mRS definition instead of a dichotomous one. Results of random-effects, inverse variance–weighted meta-analysis confirmed our previous results, showing no significant difference in functional outcomes after intensive BP reduction for both patients with deep ICH

(eFigure 3, upper panel) and lobar ICH (eFigure 4, lower panel). In our third sensitivity analysis, we repeated steps 1 and 2 using unadjusted results from ATACH-2. Findings were consistent with the primary analyses, showing no significant difference in functional outcomes after intensive BP reduction across all models (eFigure 4).

Risk of Bias of Included Trials

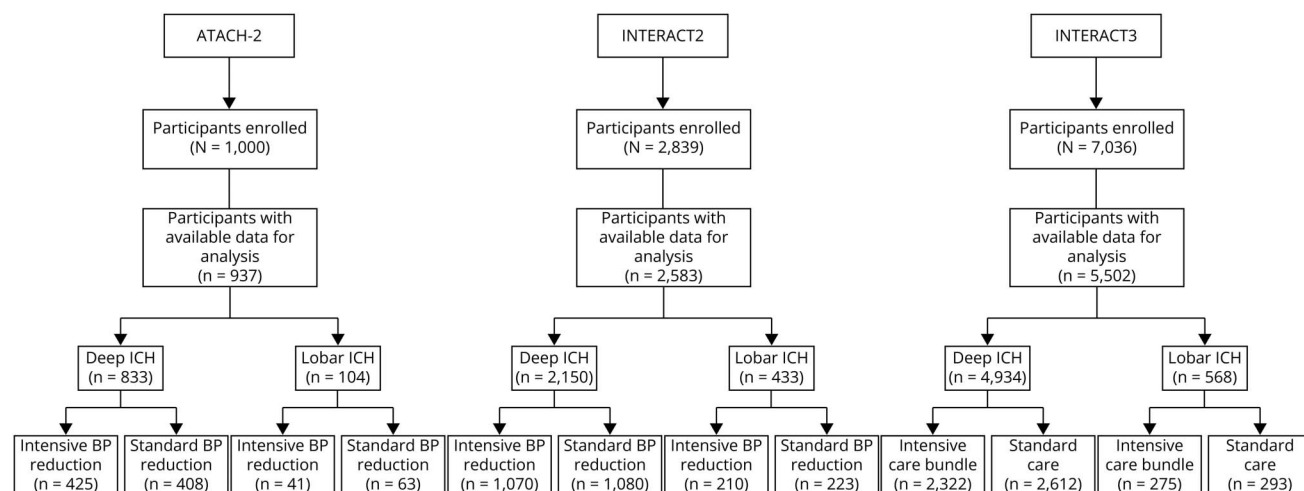
Overall, INTERACT2 and INTERACT3 were judged to have low risk of bias across all domains (eFigure 5). ATACH-2 had some concerns for D1 (bias arising from the randomization process) and D3 (bias due to missing outcome data), mainly because of limited reporting on allocation concealment and handling of missing outcomes. The remaining domains in ATACH-2 were considered to be at low risk. None of the 3 trials was judged to be at high risk of bias in any domain.

Discussion

In this stepwise meta-analysis of the landmark ATACH-2, INTERACT2, and INTERACT3 randomized clinical trials, we found that functional outcomes after intensive BP reduction did not differ significantly according to ICH location. These results remained consistent across most sensitivity analyses, including the use of ordinal functional outcome scales and unadjusted pooled results for ATACH-2. However, sensitivity analyses using fixed-effect estimates found a signal suggesting benefit of intensive BP reduction among patients with deep ICH in contrast to those with lobar ICH.

Previous research has shown that intensive BP reduction reduces the risk of perihematomal edema and hematoma

Figure 1 Flowchart Depicting the Number of Patients With Lobar and Deep Intracerebral Hemorrhage Enrolled, With Data Available for Analysis, and Randomly Assigned to Either the Intervention or Nonintervention Arms in the ATACH-2, INTERACT-2, and INTERACT3 Randomized Clinical Trials



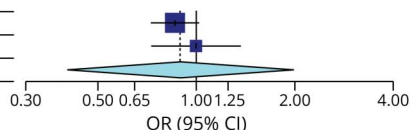
ATACH-2 = Antihypertensive Treatment of Acute Cerebral Hemorrhage-2; BP = blood pressure; ICH = intracerebral hemorrhage; INTERACT2 = Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 2; INTERACT3 = Intensive Care Bundle with Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 3.

Figure 2 Forest Plot Showing the Step 1 and Step 2 Results for the Pooled Effect of Intensive Blood Pressure Reduction on Functional Outcomes for the ATACH-2, INTERACT2, and INTERACT3 Randomized Clinical Trials

Step 1: ATACH-2 and INTERACT2

Deep ICH

Study	OR (95% CI)
INTERACT2	0.86 (0.73–1.02)
ATACH-2	1.00 (0.73–1.36)
Total	0.89 (0.40–1.98)

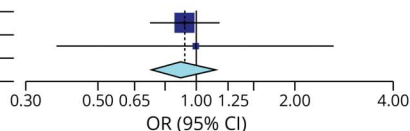


Heterogeneity: $\chi^2_1 = 0.70$ ($p = 0.40$), $I^2 = 0\%$

Test for overall effect: $t_1 = -1.85$ ($p = 0.31$)

Lobar ICH

Study	OR (95% CI)
INTERACT2	0.92 (0.72–1.17)
ATACH-2	1.00 (0.38–2.63)
Total	0.92 (0.73–1.17)



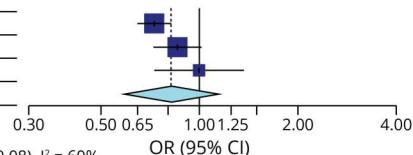
Heterogeneity: $\chi^2_1 = 0.02$ ($p = 0.88$), $I^2 = 0\%$

Test for overall effect: $t_1 = -4.27$ ($p = 0.15$)

Step 2: ATACH-2, INTERACT2, and INTERACT3

Deep ICH

Study	OR (95% CI)
INTERACT3	0.73 (0.65–0.82)
INTERACT2	0.86 (0.73–1.02)
ATACH-2	1.00 (0.73–1.36)
Total	0.82 (0.57–1.18)

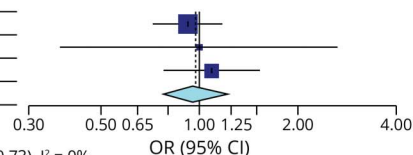


Heterogeneity: $\chi^2_2 = 4.94$ ($p = 0.08$), $I^2 = 60\%$

Test for overall effect: $t_2 = -2.32$ ($p = 0.15$)

Lobar ICH

Study	OR (95% CI)
INTERACT2	0.92 (0.72–1.17)
ATACH-2	1.00 (0.38–2.63)
INTERACT3	1.09 (0.77–1.53)
Total	0.97 (0.77–1.23)



Heterogeneity: $\chi^2_2 = 0.63$ ($p = 0.73$), $I^2 = 0\%$

Test for overall effect: $t_2 = -0.48$ ($p = 0.68$)

ATACH-2 = Antihypertensive Treatment of Acute Cerebral Hemorrhage-2; INTERACT2 = Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 2; INTERACT3 = Intensive Care Bundle with Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 3; OR = odds ratio.

expansion, the latter being one of the strongest independent predictors of poor functional outcomes in patients with deep ICH.^{11,16,24,25} Our analysis builds on this evidence by examining functional outcomes stratified by hematoma location, guided by biological considerations supporting the hypothesis that deep ICH might exhibit preferential responsiveness to intensive BP reduction. First, deep ICH is strongly associated with chronic hypertension and hypertensive vasculopathy of deep penetrating arteries.¹² Second, even small hemorrhages in deep locations are clinically devastating because of their proximity to eloquent brain structures.^{14,15} Third, hematoma expansion is more frequent in deep ICH.¹³ Thus, reducing the risk of hematoma expansion and perihematomal edema by treating the primary pathogenic mechanism could preferentially benefit patients with deep ICH. It is important to note that recent evidence indicates no association between intensive BP lowering and subacute ischemic brain lesions on diffusion-weighted MRI, addressing previous safety concerns and supporting the plausibility of benefit in patients with chronic hypertensive arteriopathy and deep ICH.²⁶

Although high BP is also present in patients with acute lobar ICH, it is not as prevalent as in patients with deep ICH, it is less severe, and its pathogenic role is less clear.¹¹ Lobar ICH is etiologically more heterogeneous than deep ICH, but cerebral amyloid angiopathy is arguably its most frequent and important mechanism.²⁷ Patients with lobar ICH are also clinically distinct from those with deep ICH, being usually older, having a higher prevalence of dementia, and developing renal complications during hospitalization more frequently.^{28,29} In addition, the benefit of early minimally invasive hematoma evacuation restricted to lobar ICH in the Early Minimally

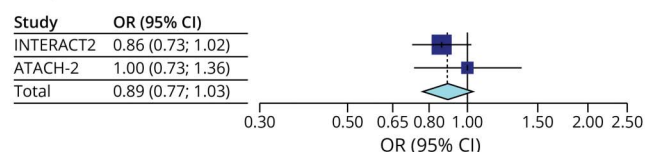
Invasive Removal of Intracerebral Hemorrhage trial can be viewed as evidence of biologic differences among ICH subtypes, or alternatively as differences related to the location of neuronal lesions, both of which support the rationale for tailored management according to hematoma location.³⁰ Taken together, these features suggest a biologically plausible framework for differential treatment effects for deep and lobar ICH.

Despite the well-described biological differences between deep and lobar ICH, our analysis did not yield convincing evidence that the effect of intensive BP reduction differs by hematoma location. Several clinical and methodological sources of heterogeneity across the trials could account, at least in part, for these null findings. These include differences in the timing and definition of functional outcomes, variations in the trial interventions, and differences in patient demographics. For example, ATACH-2 and INTERACT2 assessed 3-month mRS outcomes, whereas INTERACT3 used 6-month mRS scores; good outcome was defined as mRS scores 0–3 in ATACH-2 and INTERACT3, but as mRS scores 0–2 in INTERACT2. In addition, INTERACT3 evaluated a multifaceted care bundle rather than isolated BP reduction. Finally, Asian participants were overrepresented in all 3 trials, comprising roughly 56% of ATACH-2,⁶ 68% of INTERACT2,⁵ and as high as 92% of INTERACT3,⁷ further contributing to between-trial variability. We addressed these sources of heterogeneity in several ways. First, we structured the meta-analysis in 2 steps, with step 1 combining only ATACH-2 and INTERACT2, which were more similar in design and outcome timing and step 2 adding INTERACT3 to incorporate its larger sample size. Although INTERACT3

Figure 3 Forest Plot Showing Sensitivity Analysis Results for the Pooled Effect of Intensive Blood Pressure Reduction on Functional Outcomes Ascertained Using Fixed-Effects Inverse Variance–Weighted Meta-Analysis for the ATACH-2, INTERACT2, and INTERACT3 Trials

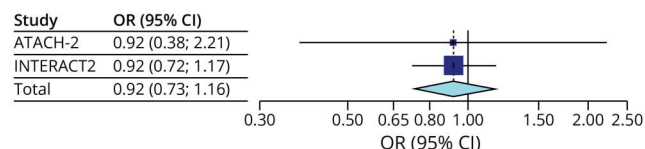
Step 1: ATACH-2 and INTERACT2

Deep ICH



Heterogeneity: $\chi^2_1 = 0.70$ ($p = 0.40$), $I^2 = 0\%$

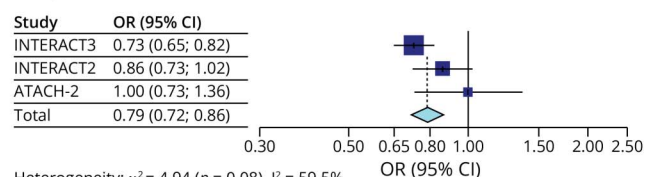
Lobar ICH



Heterogeneity: $\chi^2_1 = 0.00$ ($p > 0.99$), $I^2 = 0.0\%$

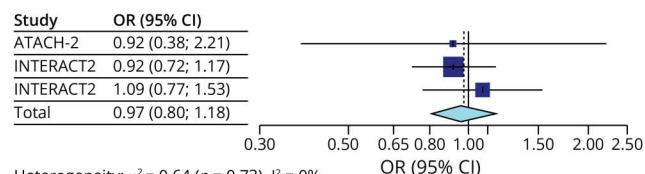
Step 2: ATACH-2, INTERACT2, and INTERACT3

Deep ICH



Heterogeneity: $\chi^2_2 = 4.94$ ($p = 0.08$), $I^2 = 59.5\%$

Lobar ICH



Heterogeneity: $\chi^2_2 = 0.64$ ($p = 0.73$), $I^2 = 0\%$

ATACH-2 = Antihypertensive Treatment of Acute Cerebral Hemorrhage-2; INTERACT2 = Intensive Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 2; INTERACT3 = Intensive Care Bundle with Blood Pressure Reduction in Acute Cerebral Hemorrhage Trial 3; OR = odds ratio.

used a care bundle, intensive BP lowering remained its central component, as 90% of participants presented with systolic BP > 140 mm Hg and only a minority met criteria for the other bundle elements (hyperglycemia 35%, fever 1.7%, elevated international normalized ratio 1.2%).⁷ Second, although formal heterogeneity estimates ($I^2 < 70\%$) did not indicate high variability, we used random-effects models for all pooled estimates to provide more conservative effect sizes and reduce the likelihood of spurious associations. Third, we conducted a sensitivity analysis using ordinal mRS data from ATACH-2 and INTERACT3 to evaluate whether conclusions differed when using a more granular functional outcome scale. Notably, an additional sensitivity analysis using fixed-effects models, which do not account for interstudy heterogeneity, yielded a statistically significant reduction of 21% in poor outcome for deep ICH. This signal, although interpreted cautiously because of methodological heterogeneity as described above, is consistent with biological expectations and highlights the need for future trials specifically designed to evaluate effect modification by ICH location.

Several additional considerations are important when interpreting these findings. Although the direction of effect in deep ICH during step 2 was consistent with biological expectations, the dominance of INTERACT3 in the pooled estimate warrants caution. As the largest of the 3 trials and the major contributor of deep ICH cases, its particular features, including the inclusion of larger hematoma volumes that are more likely to respond to BP lowering, the use of a treatment bundle rather than BP reduction alone, and the assessment of outcomes at 6 months, may influence the observed results. Moreover, the need for such a large sample to detect outcome differences with BP lowering suggests that, if present, the

clinical benefit is modest at best. Second, the proportion of Asian participants was substantially higher in INTERACT2 and INTERACT3 than in ATACH-2, reflecting regional differences in recruitment. Whether these demographic differences modify treatment response is uncertain; however, hypertension-related deep ICH is more prevalent in Asian individuals, which could potentially influence the observed effects.^{31,32} Third, INTERACT3 participants had somewhat larger median hematoma volumes and slightly higher rates of intraventricular hemorrhage than those in ATACH-2 and INTERACT2, both important predictors of poor outcome.⁷ In addition, given that the risk of hematoma expansion increases with ICH volumes, the larger hematomas enrolled in INTERACT3 likely created a higher risk cohort more susceptible to the beneficial effects of BP lowering.³³ These imbalances may contribute additional variability to pooled results. However, the consistent lack of statistical significance across multiple sensitivity analyses suggests that demographic and clinical heterogeneity alone is unlikely to fully explain the null findings.

Comprehensive individual patient data meta-analyses pooling all landmark ICH trials are needed to overcome the statistical limitations of this study and more robustly evaluate whether the effect of intensive BP reduction differs between deep and lobar ICH. Such evidence could help refine current approaches and potentially guide the development of more tailored management strategies that consider hematoma location and its underlying biological determinants. In a hypothetical future model, once ICH and its location are identified on head CT in patients presenting with acute neurologic symptoms, location-specific clinical pathways might be activated. For deep ICH, this could involve expedited initiation of

IV antihypertensive therapy with close monitoring of BP and renal function,^{11,16} whereas for lobar ICH, emphasis might shift toward urgent neurosurgical consultation to assess eligibility for minimally invasive hematoma evacuation^{30,34} alongside careful (but not overly aggressive) BP management. It is important to note that the *speed* of therapy initiation may itself be a critical determinant of outcome for ICH: recent evidence demonstrates that shorter antihypertensive treatment times are associated with better discharge outcomes in patients with ICH, yet current acute treatment strategies are markedly slower for ICH than for ischemic stroke.³⁵ These speculative considerations underscore the potential for location-informed and time-sensitive treatment paradigms, but such approaches remain hypothesis-generating and require rigorous validation before informing clinical guidelines, especially given the null results reported in this study.

Our study strengths are its large sample size, which enabled us to analyze a large number of patients with lobar ICHs, a trait systematically underrepresented in clinical trials of ICH,¹¹ and the use of previously validated clinical trial data. Several limitations should also be acknowledged. First, the trials differed in their design, outcome timing, and functional outcome definitions, which introduced methodological heterogeneity. Although we addressed these differences by structuring the meta-analysis in 2 steps and using random-effects models for all pooled estimates, these discrepancies remain important when interpreting the findings. Second, our primary analyses relied on dichotomous mRS cut points, which are less sensitive than ordinal analyses.³⁶ However, our ordinal sensitivity analyses yielded consistent results, suggesting that our conclusions were robust to outcome definition. Third, INTERACT3 implemented a multifaceted care bundle rather than isolated BP control. Nevertheless, BP lowering was the dominant component and accounted for most of the bundle's beneficial effect.¹⁷ The inclusion of patients with anticoagulation-associated ICH could theoretically confound treatment estimates, given their worse prognosis. However, only 83 such patients were enrolled (1.2%),⁷ and only 2 involved direct oral anticoagulants, indicating that anticoagulation reversal likely exerted minimal influence on the pooled effect. Because the study did not present location-specific data for anticoagulation-associated ICH, it is useful to note that meta-analytic evidence shows no clear predilection for deep or lobar locations.^{7,8,37,38}

In this meta-analysis of randomized trials evaluating acute BP management in ICH, intensive BP reduction did not show statistically significant differential effects between deep and lobar ICH. Although mechanistic considerations and effect directionality suggest the possibility of benefit in deep ICH, these results remain hypothesis-generating and should be interpreted cautiously. Future well-powered trials and meta-analyses including individual-patient data, specifically designed to test treatment-location interactions, are needed to determine whether tailored BP management strategies could improve outcomes across ICH subtypes.

Author Contributions

S. Clocchiatti-Tuozzo: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. A.I. Qureshi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. C.A. Rivier: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Huo: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. C. Matouk: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Payabvash: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S.B. Murthy: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. K.N. Sheth: drafting/revision of the manuscript for content, including medical writing for content. A.A. Rabinstein: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. G.J. Falcone: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M.A. Hawkes: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; Analysis or interpretation of data.

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